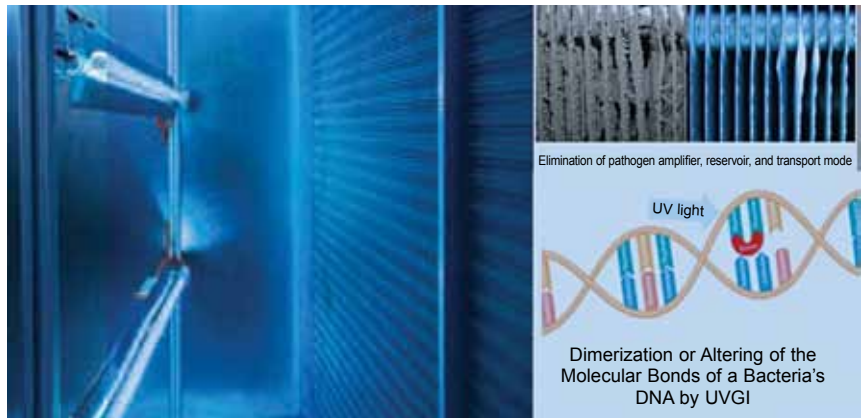


floor. These are blown into the air by human movement from one room to another. Lower-air room disinfectants can be used to decontaminate the floors along with the walls uniformly to maintain them in sterile condition and prevent the spread of virus.

Such devices are mounted on a wall at a height of 450 to 600mm above ground level. This has less risk of human exposure and wearing normal full-length pants and footwear can protect occupants further. These are usually recommended for corridors leading to sterile zones.

Upper-air disinfectants. These have proven to be the most effective solution for controlling airborne infections including coronavirus. Remember that due to the large volume of air, it is not possible for most HVACs or centralised AC units to refresh air every five minutes. And, as the size of space increases, so does the inadequacy of the centralised units.

According to the CDC, “UV research on SARS-CoV-2 indicates that the virus that causes Covid-19 is very similar to other coronaviruses (for example, SARS and MERS) regarding the UV dose necessary to inactivate it. Upper-room UVGI systems can be used to control SARS-CoV-2 as a useful ventilation



UVGI for air handling units (Courtesy: K Clean Services)

tool to consider for reducing the spread of infectious pathogens.”

Upper-room disinfectants tackle the challenge of refreshing the air quickly by working on a room-wise basis. Frankly, they seem to deploy many of the common-sense principles that have been in existence for quite some time now, and hence no one holds a patent to this tech either.

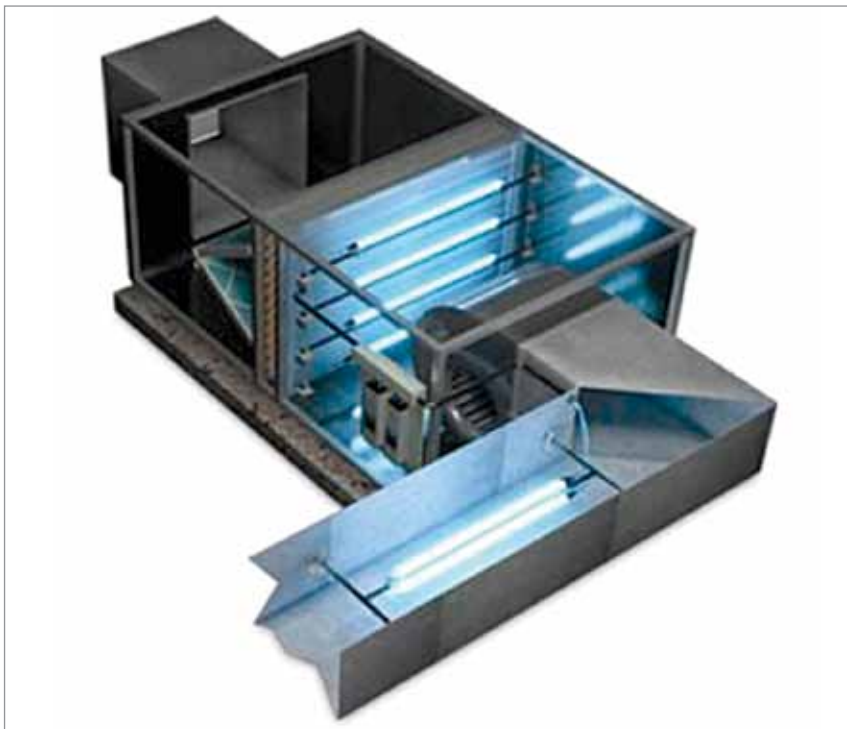
They work on the principle that hot air, when you exhale, rises before cooling and settling down. Thus, by placing UV luminary devices at a height just above our heads they eliminate pathogens as soon as one exhales, thereby eliminating chances

of transmission. And some units come with suction blowers which also suck the air upwards towards the unit so that speed and effectiveness of decontamination increases, thereby further reducing the viral load.

Since they are able to disinfect air closer to and above people in a room, they can be deployed even in high-risk indoor settings where adequate natural ventilation is not possible. “To minimise UVC exposure, upper-air disinfection solutions could be used without significant risk for long-term effects such as skin cancer. Upper-air UVC device construction and placement prevents excessive UV energy from striking occupants below,” adds Hina Gupta.

UV chambers. UVGI chambers are commonly used for sterilisation of medical equipment, electronic instruments, PPE masks, and other small objects. These typically come with a timer to manage the period of exposure to UV. For effectiveness, the chamber must be designed in such a way that UVC light falls all around the object placed inside. Precautions should be taken to place objects with adequate space between them to avoid overlaying. The UVC light must be shut down completely before opening the chamber and care should be taken if the lamp inside emits ozone.

UV doors and corridors. UVC radiation cannot be used directly on humans due to its harmful nature. However, experts are working on designing UVGI walk-through corridors with a safe level of 222nm UVC, especially for personnel treating Covid-19 patients. These will allow reuse of the PPE kit, as after decon-



Installation of in-duct UVC solutions (Credit: Binay Opto Electronics)